

Unit 8: Special Equations and Classical Models

Prepared by Staples Education

Condensed Cheat Sheet

Unit 8 at a Glance: Core Sub-Topics

Unit 8 moves from the algebraic theory of first-order ODEs into the classical applied models that drive physics, biology, and economics. It opens with two *special equation types*—the Clairaut equation (singular-solution envelopes) and pursuit/tractrix curves—before building three major applied models: the Navier–Stokes-derived plane Poiseuille velocity profile, the supply-and-demand price-adjustment ODE, and the Lotka–Volterra predator-prey and competition systems. The unit closes with a structured survey of first-order solution methods.

- **8.1 The Clairaut Equation and Singular Solutions** — general solution $y = Cx + f(C)$; singular (envelope) solution from the system $\{y = xp + f(p), 0 = x + f'(p)\}$.
- **8.2 Pursuit Curves and the Tractrix** — tractrix DE $dy/dx = -\sqrt{a^2 - x^2}/x$; general pursuit $xy'' = k\sqrt{1 + (y')^2}$ with speed ratio $k = v_m/v_d$.
- **8.3 Fluid Dynamics and Viscosity** — Newtonian shear $\tau = \mu du/dy$; plane Poiseuille flow $\mu u'' = -G$ with no-slip boundary conditions.
- **8.4 Economic Models, Supply, Demand, and Price** — price ODE $dp/dt = k(D - S)$; equilibrium $p^* = (a + c)/(b + d)$; convergence for $k(b + d) > 0$.
- **8.5 The Lotka–Volterra Predator–Prey Model** — coexistence equilibrium $(c/d, a/b)$; conserved quantity $H = dx - c \ln x + by - a \ln y$.
- **8.6 Competition Models** — two-species carrying-capacity system; nullcline inequalities for stable coexistence versus competitive exclusion.
- **8.7 A Survey of Solution Methods** — decision tree: separable, linear (integrating factor), Bernoulli, exact; classify by structure before solving.

Part I — Condensed Cheat Sheet

8.1 The Clairaut Equation and Singular Solutions

A *Clairaut equation* has the special form $y = xy' + f(y')$. Setting $p = y'$ and differentiating both sides with respect to x yields $(x + f'(p)) dp/dx = 0$, splitting into two branches whose solutions are the general family of straight lines and the singular envelope.

Clairaut Equation: General and Singular Solutions

For $y = xp + f(p)$ with $p = y'$:

General solution: $y = Cx + f(C)$ ($p = C = \text{const}$),

Singular solution (envelope):
$$\begin{cases} y = xp + f(p) \\ 0 = x + f'(p) \end{cases}$$

Eliminate p from this system to obtain the envelope: a curve tangent to every line in the general family. The singular solution satisfies the ODE but *cannot* be obtained from the general solution by any choice of C .

Envelope Elimination Procedure

1. Write the equation as $F(x, y, C) = y - Cx - f(C) = 0$.
2. Differentiate with respect to C : $\partial F / \partial C = -x - f'(C) = 0$, giving $x = -f'(C)$.
3. Solve for $C = C(x)$ from step 2 (if invertible) and substitute back into $F = 0$ to obtain the envelope curve $y = y(x)$.
4. Verify: the envelope must satisfy the original ODE directly.

Common Pitfall: Singular Solution Is Not a Particular Solution

The singular solution is *not* a member of the general family $y = Cx + f(C)$ for any finite C . Do not try to match it by plugging a value of C into the general formula; instead, derive it by the envelope elimination system. Attempting to recover it as a special case of the general solution will always fail.

8.2 Pursuit Curves and the Tractrix

A *pursuit curve* traces the path of a pursuer who always aims directly at a moving target. The tractrix is the classical limiting case in which the pursuer is dragged at constant distance a from a target walking along a line.

Tractrix Differential Equation and Solution

Target on the y -axis from $(0, 0)$, pursuer starts at $(a, 0)$, equal speeds, string length a :

$$\frac{dy}{dx} = -\frac{\sqrt{a^2 - x^2}}{x}, \quad 0 < x \leq a, \quad y(a) = 0.$$

Solution (via trigonometric substitution $x = a \sin \theta$):

$$y(x) = a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right) - \sqrt{a^2 - x^2}.$$

General Pursuit Equation

Dog at origin pursues owner on line $x = c$ moving at speed v_m ; dog speed $v_d > v_m$, always aimed at owner. Speed ratio $k = v_m/v_d < 1$:

$$x y'' = k \sqrt{1 + (y')^2}, \quad y(c) = 0, \quad y'(c) = 0.$$

Set $p = y'$, separate: $dp/\sqrt{1 + p^2} = k dx/x$. After solving and back-substituting ($k \neq 1$):

$$y(x) = \frac{1}{2} \left[\frac{x^{k+1}}{(k+1)c^k} - \frac{c^k x^{1-k}}{1-k} \right] + \frac{ck}{1-k^2}.$$

The dog captures the owner at height $y(0) = ck/(1 - k^2)$.

Common Pitfall: Direction of the Derivative Sign

In the tractrix the slope dy/dx is *negative* because x decreases as the pursuer is dragged toward the y -axis. Always verify the geometric sign: the string pulls the pursuer toward the target, so the y -component of motion is the signed vertical projection of the unit tangent to the path.

8.3 Fluid Dynamics and Viscosity

Newtonian viscosity relates shear stress τ to the velocity gradient: $\tau = \mu du/dy$. In plane Poiseuille flow between parallel plates the momentum balance reduces to a simple second-order constant-coefficient ODE whose solution is a parabolic profile.

Plane Poiseuille Velocity Profile

Flow between plates $y = 0$ and $y = h$ driven by pressure gradient $G > 0$; viscosity μ ; no-slip: $u(0) = u(h) = 0$:

$$\mu u''(y) = -G \quad \implies \quad u(y) = \frac{G}{2\mu} y(h - y).$$

Maximum velocity at the centerline:

$$y_{\max} = \frac{h}{2}, \quad u_{\max} = \frac{Gh^2}{8\mu}.$$

Shear stress: $\tau(y) = \mu u'(y) = \frac{G}{2}(h - 2y)$, linear, zero at the centerline and $\pm Gh/2$ at the walls.

2D Incompressible Continuity and Stream Function

For steady incompressible 2D flow, mass conservation requires

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Given u , integrate $\partial v/\partial y = -\partial u/\partial x$ in y to find the most general $v(x, y) = -\int (\partial u/\partial x) dy + g(x)$. Streamlines satisfy $dy/dx = v/u$, or equivalently the stream function $\psi(x, y) = C$ where $u = \partial\psi/\partial y$ and $v = -\partial\psi/\partial x$.

Common Pitfall: Applying No-Slip at Both Walls

The Poiseuille ODE $\mu u'' = -G$ integrates to a quadratic with *two* free constants. Both boundary conditions ($u = 0$ at $y = 0$ and $y = h$) are required to pin them. Using only one wall gives a family of parabolas, not the physical velocity profile.

8.4 Economic Models, Supply, Demand, and Price

A linear price-adjustment model assumes price changes at a rate proportional to excess demand. With linear demand and supply the governing ODE is first-order linear with constant coefficients, and the price converges exponentially to the market-clearing equilibrium.

Price Adjustment ODE and Equilibrium

Demand $D = a - bp$, supply $S = -c + dp$ ($a, b, c, d > 0$). Walrasian adjustment:

$$\frac{dp}{dt} = k(D - S) = k[(a + c) - (b + d)p].$$

Solution with $p(0) = p_0$:

$$p(t) = p^* + (p_0 - p^*)e^{-k(b+d)t}, \quad p^* = \frac{a + c}{b + d}.$$

Convergence: $k(b + d) > 0$ always holds, so $p(t) \rightarrow p^*$ monotonically for any initial price p_0 .

Solving the Price ODE via Integrating Factor

1. Write $dp/dt + k(b + d)p = k(a + c)$ (linear, constant coefficients).
2. Integrating factor: $\mu(t) = e^{k(b+d)t}$.
3. Multiply and integrate: $p(t) = k(a + c)/[k(b + d)] + Ce^{-k(b+d)t}$.
4. Apply $p(0) = p_0$: $C = p_0 - p^*$.

8.5 The Lotka–Volterra Predator–Prey Model

The Lotka–Volterra system models cyclic predator–prey oscillations. Its trajectories are *closed orbits* in the phase plane, guaranteed by a first integral (conserved quantity) $H(x, y)$ that is constant on every solution curve.

Lotka–Volterra System and Conserved Quantity

Prey x , predator y ; parameters $a, b, c, d > 0$:

$$\dot{x} = ax - bxy, \quad \dot{y} = -cy + dxy.$$

Equilibria: $(0, 0)$ (extinction) and $(x^*, y^*) = (c/d, a/b)$ (coexistence).

Divide $\dot{y}/\dot{x}$ to get dy/dx and separate:

$$\frac{a - by}{y} dy = \frac{-c + dx}{x} dx \implies H(x, y) = dx - c \ln x + by - a \ln y = \text{const.}$$

Level curves of H are closed orbits; the coexistence equilibrium is the unique minimum of H for $x, y > 0$.

Finding Equilibria and the Conserved Quantity

1. Set $\dot{x} = 0$: $x(a - by) = 0 \Rightarrow x = 0$ or $y = a/b$.
2. Set $\dot{y} = 0$: $y(-c + dx) = 0 \Rightarrow y = 0$ or $x = c/d$.
3. Form $dy/dx = \dot{y}/\dot{x}$; cross-multiply to separate $(a/y - b) dy = (-c/x + d) dx$.
4. Integrate both sides; rearrange to $H(x, y) = \text{const.}$
5. Verify H is minimal at $(c/d, a/b)$ by computing $\partial H/\partial x$ and $\partial H/\partial y$.

Common Pitfall: Confusing Equilibrium Stability

The coexistence equilibrium $(c/d, a/b)$ is a *center* in the Lotka–Volterra model, not a stable spiral. Trajectories orbit it periodically but never actually reach it. Adding even small prey-growth saturation converts the center to a stable spiral; the pure Lotka–Volterra centers are structurally unstable.

8.6 Competition Models

Two-species competition extends the logistic model by letting each species reduce the other's effective carrying capacity. The outcome—coexistence or competitive exclusion—depends entirely on the relative positions of the x - and y -nullclines.

Two-Species Competition System and Nullclines

$$\dot{x} = r_1 x \left(1 - \frac{x + \alpha y}{K_1} \right), \quad \dot{y} = r_2 y \left(1 - \frac{y + \beta x}{K_2} \right).$$

Four equilibria: $(0, 0)$, $(K_1, 0)$, $(0, K_2)$, and the interior point (x^*, y^*) obtained by solving $x + \alpha y = K_1$ and $\beta x + y = K_2$ simultaneously:

$$x^* = \frac{K_1 - \alpha K_2}{1 - \alpha\beta}, \quad y^* = \frac{K_2 - \beta K_1}{1 - \alpha\beta}.$$

x -nullcline (non-trivial): $x + \alpha y = K_1$

y -nullcline (non-trivial): $\beta x + y = K_2$.

Coexistence vs. Competitive Exclusion: Nullcline Intercept Test

Compare the nullcline intercepts on each axis:

- **Stable coexistence** (E_3 stable node, E_1 and E_2 saddles):

$$K_1 < \frac{K_2}{\beta} \quad \text{and} \quad K_2 < \frac{K_1}{\alpha} \quad (\text{equivalently } \alpha\beta < 1).$$

- **Bistable exclusion** (E_3 unstable saddle, E_1 and E_2 stable nodes):

$$K_1 > \frac{K_2}{\beta} \quad \text{and} \quad K_2 > \frac{K_1}{\alpha} \quad (\text{equivalently } \alpha\beta > 1).$$

- **Species x always wins** (E_1 globally stable): $K_1 > K_2/\beta$ and $K_2 < K_1/\alpha$.
- **Species y always wins** (E_2 globally stable): $K_1 < K_2/\beta$ and $K_2 > K_1/\alpha$.

Common Pitfall: Interior Equilibrium Feasibility

The interior equilibrium $E_3 = (x^*, y^*)$ is biologically meaningful only when *both* $x^* > 0$ and $y^* > 0$. Check the signs of the numerators *and* denominator of the formulas above. When $\alpha\beta = 1$ the two nullclines are parallel (no interior intersection); when $\alpha\beta > 1$ the intersection exists but is in the positive quadrant yet is unstable—it does not represent a viable long-run coexistence state.

Unit 8 Key Takeaway

Clairaut equations split into a **general line family** $y = Cx + f(C)$ and a **singular envelope** found by eliminating p from $\{y = xp + f(p), x + f'(p) = 0\}$. Pursuit geometry converts geometric tangency constraints directly into ODEs. The Poiseuille profile $u = (G/2\mu) y(h - y)$ and the price path $p(t) = p^* + (p_0 - p^*)e^{-k(b+d)t}$ both follow from first- or second-order linear ODEs with boundary data. In Lotka–Volterra dynamics, closed orbits are level sets of $H = dx - c \ln x + by - a \ln y$; in competition models the outcome—**coexistence or exclusion**—is decided by whether $\alpha\beta < 1$ or $\alpha\beta > 1$.